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DUNMAN HIGH SCHOOL **Preliminary Examination** **Year 6**

H2 BIOLOGY

9648/02

Paper 2 Structured and Free-Response Questions

15 September 2014

2 hours

Additional Materials: Writing paper

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Section B Structured Questions

Answer **all** questions.

Write your answers on space provided in the Question Paper.

Section C Free-Response Questions

Answer **one** question. Your answer to Section C must be in continuous prose, where appropriate. Write your answers on the writing paper provided.

Submit your answers to Sections B and Section C separately.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

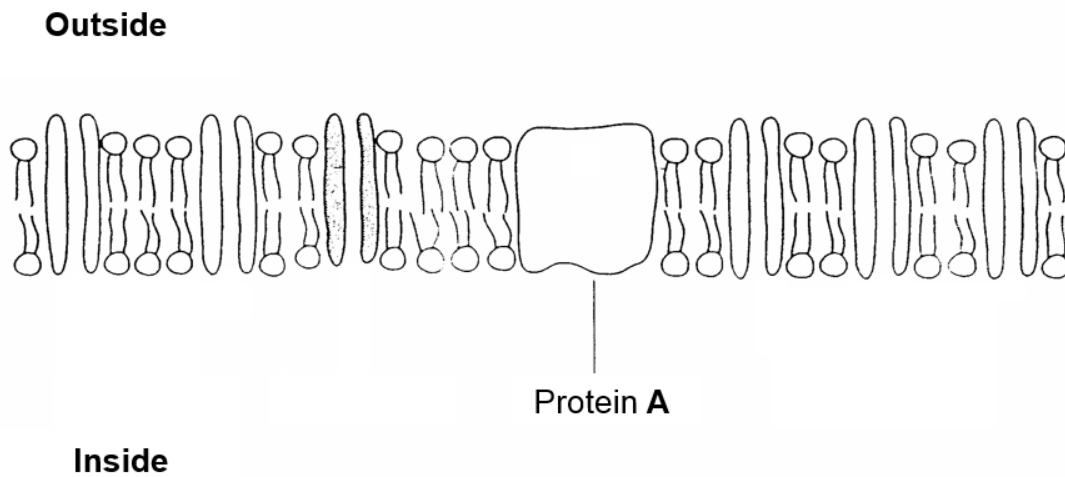
For Examiner's Use	
Section A [40]	
Section B [80]	
1	/ 14
2	/ 5
3	/ 10
4	/ 10
5	/ 8
6	/ 13
7	/ 10
8	/ 10
Section C [20]	
1 / 2	
Total [140]	

This document consists of **23** printed pages and **1** blank page.

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Section B: Structured Questions (80 marks)Answer **all** questions in this section.*For
Examiner's
use***Question 1**

- (a) **Fig 1.1** is a diagram representing a section of a membrane. It shows different proteins that are involved in maintaining an unequal distribution of ions inside and outside the cell.

**Fig1.1**

Protein A requires ATP to function. Describe the role of protein A.[2]

- (b) **Fig1.2** shows an electron micrograph of a cell from the lung that is specialised to take up material by endocytosis. Darkly staining vesicles containing inorganic aluminiumsilicate crystals are labelled in **Fig1.2**. These crystals are only ever seen in such cells in the lungs of cigarette smokers. Aluminium silicate is known to be present in tobacco smoke.

*For
Examiner's
use*



Fig1.2

- (i) Explain how endocytosis allows foreign material such as aluminium silicate to be taken up by cells. [2]

- (ii) Describe how foreign material is normally dealt with once in these vesicles. [2]

- (c) The enzyme catalase catalyses the decomposition of hydrogen peroxide to water and oxygen. The rate of reaction of catalase can be determined by plotting the volume of oxygen produced against time.

For
Examiner's
use

Fig 1.3 below shows the graph of formation of oxygen production against time for catalase at two different temperatures.

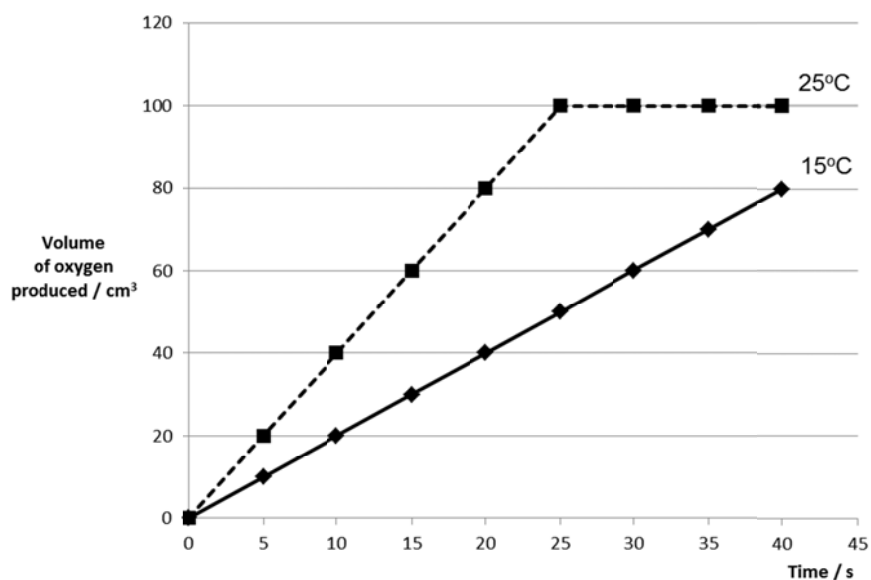


Fig 1.3

With reference to **Fig 1.3**, describe and explain the effects of temperature on the rate of catalase. [4]

- (d) HMG-CoA is an important key cholesterol precursor for the biosynthesis of cholesterol. **Fig 1.4** shows HMG-CoA reductase, which catalyses the rate-determining step of cholesterol biosynthesis, converting HMG-CoA to Mevalonate.

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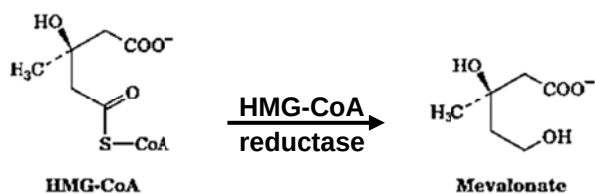


Fig 1.4

Atorvastatin (Lipitor) is an inhibitor of HMG-CoA reductase. The drug is used to treat hypercholesterolemia (high blood cholesterol).

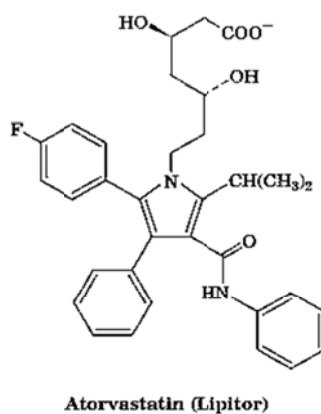


Fig 1.5

With reference to **Fig 1.4** and **1.5**, suggest how the Atorvastatin is able to inhibit HMG-CoA reductase activity. [2]

- (e) Catalase is a globular tetrameric enzyme made up of four polypeptide chains. **Fig 1.6** shows the molecular structure of catalase.

*For
Examiner's
use*

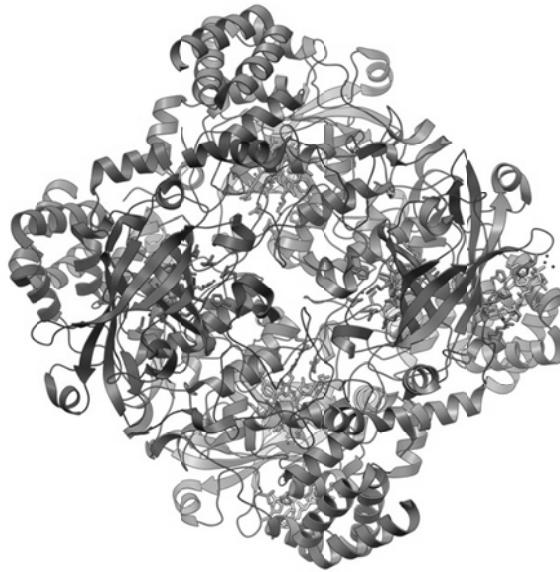


Fig 1.6

With reference to **Fig 1.6**, other than peptide bonds, briefly describe the two types of bonds that are found in catalase. [2]

Total: [14]

Question 2

Fig 2 shows some cells undergoing mitosis. Each of the cells **A**, **B**, **C** and **D** is in a different stage of mitosis.

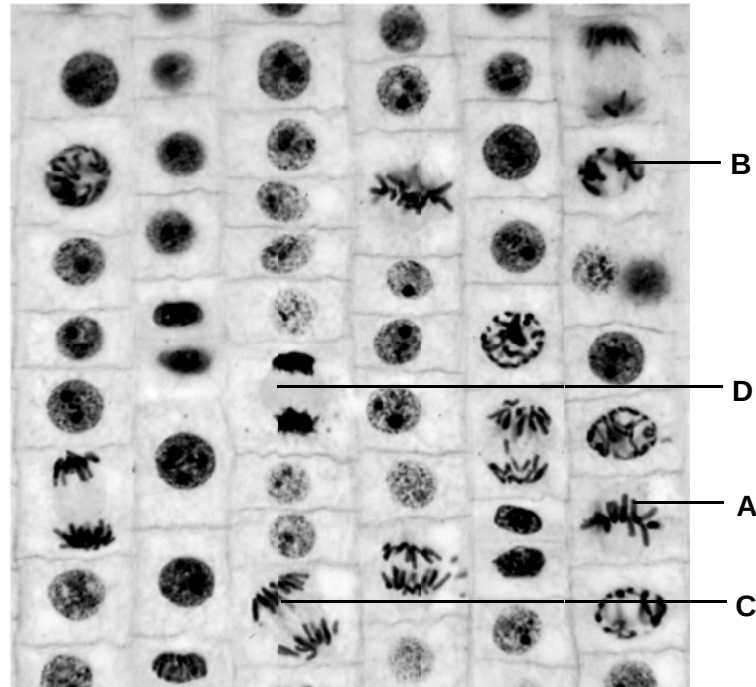


Fig 2

- (a) Using the letters provided, write the correct order of the stages in mitosis. [1]

- (b) Describe what happens to the nuclear envelope during mitosis. [2]

- (c) Paclitaxel is a drug used in the treatment of cancer. It stabilizes the microtubule polymer and protects it from disassembly during nuclear division. Explain how treatment with paclitaxel will affect the behavior of chromosomes during nuclear division and how it serves as an effective treatment. [2]

Total: [5]

Question 3

- (a) Protein synthesis in a eukaryotic cell is a multi-step process. **Fig 3.1** below shows a step during protein synthesis.

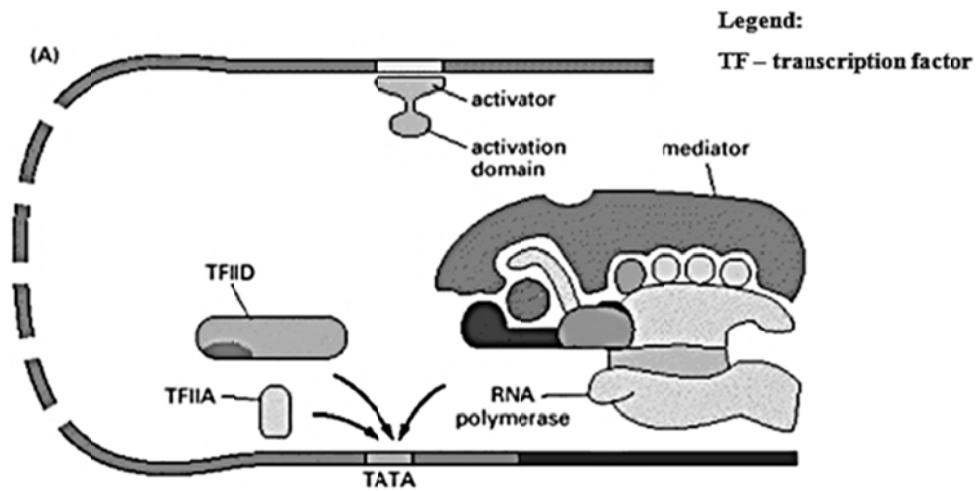


Fig 3.1

- (i) State precisely where this process can be found. [1]
- (ii) With reference to **Fig 3.1**, describe how protein synthesis will be affected when there is a mutation to the TATA region. [3]

- (b) Fig 3.2 shows the binding of RNA polymerase to DNA and the sequences of DNA which the RNA polymerase will transcribe.

For
Examiner's
use



Fig 3.2

- (i) Write down the mRNA sequence transcribed. [1]

- (ii) Explain how you arrive at your answer in (i). [2]

- (c) Describe three ways in which transcription differs from translation in protein synthesis. [3]

Total: [10]

Question 4

(a) Fig 4.1 shows the structure of an influenza virus.

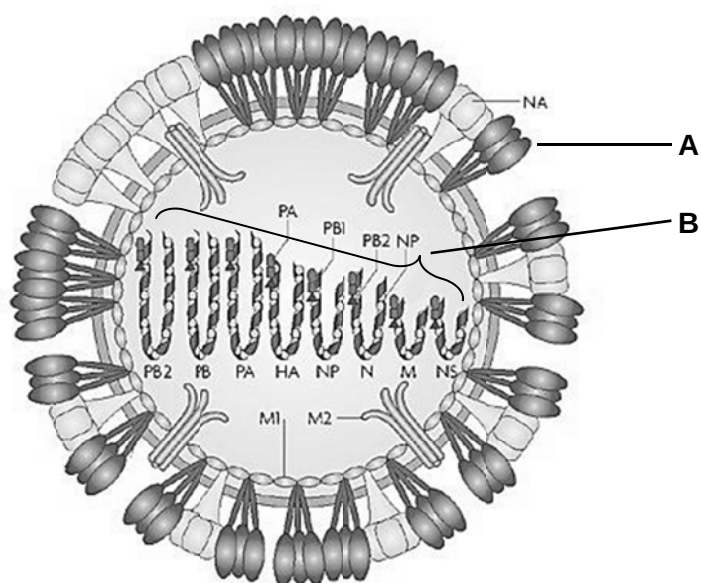


Fig 4.1

(i) Identify the molecules labelled **A** and **B**. [2]

A _____

B _____

(ii) Using named examples, explain how molecule **B** is involved in causing infections in epithelial cells. [2]

(b) Fig 4.2 shows part of the life cycle of both the influenza and HIV viruses.

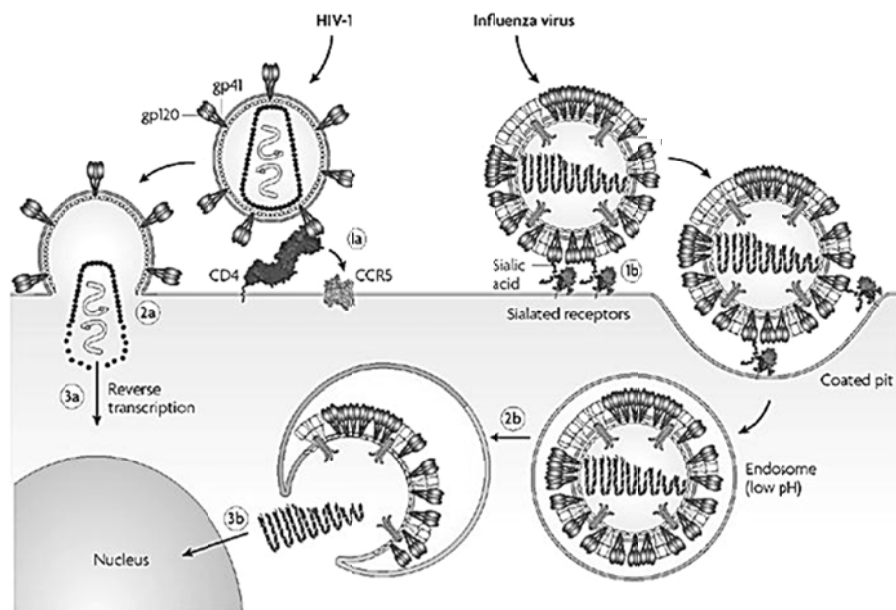


Fig 4.2

With reference to **Fig 4.2**, describe three differences in the life cycle of these two viruses. [3]

(c) Fig 4.3 is an electron micrograph showing influenza being released from the host cell.

For
Examiner's
use

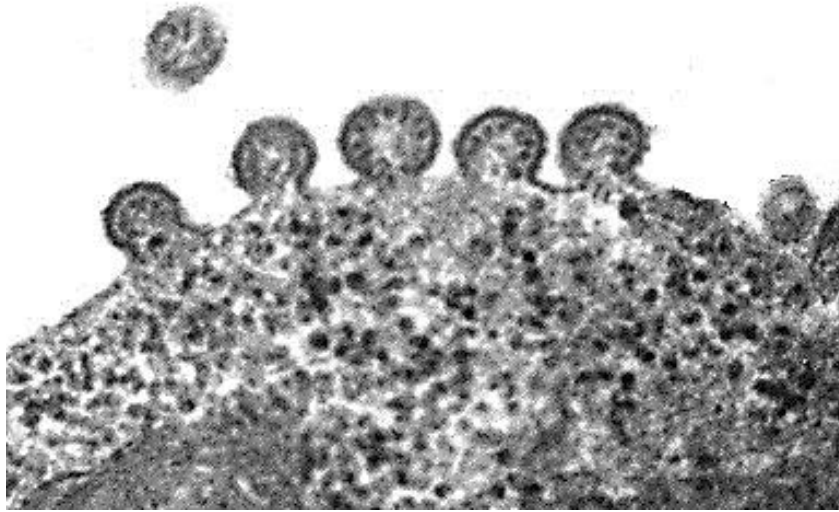


Fig 4.3

Describe how influenza acquires the outer envelope. [3]

Total: [10]

Question 5

- (a)** Compare and contrast the structure and organisation of prokaryotic and eukaryotic chromosomes. [4]

- (b)** Fig 5.1 shows the action of eIF4F and poly-adenine binding proteins (PABP). eIF4F is a translation initiation factor involved in increasing binding of ribosome to mRNA. Multiple PABP binds to the 3' polyA tail of the mRNA.

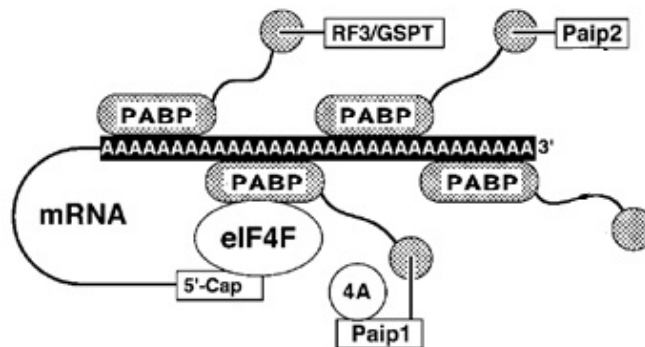


Fig 5.1

- (i) With reference to **Fig 5.1**, describe how the 5' cap increases translation. [2]

- (ii)** Suggest how PABP increases protein synthesis in a cell. [2]

Total: [8]

Question 6

Flower colour in Dahlia flowers is determined by two gene loci, A/a and B/b.

The alleles give the following phenotypes:

A	yellow	a	white
B	white	b	yellow

A cross between two white-flowered plants produces offspring of the following phenotypic ratio 3 yellow-flowered plants: 13 white flowered plants.

(a) Construct a genetic diagram to show the cross described. [3]

(b) Explain how interaction between the two genes gives rise to white-flowered plants. [2]

- (c) An inbred variety of maize, **A**, with finely striped leaves was found to have high resistance to the fungus that causes the disease, corn leaf blight.

For
Examiner's
use

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

The above F_1 generation was back-crossed to variety **A** and yielded the following results:

finely striped leaves and high resistance	80
finely striped leaves and low resistance	20
entirely green leaves and high resistance	22
entirely green leaves and low resistance	78

- (i) A chi-squared test was performed on the results of the cross to determine if the results of the test cross depart significantly from the expected ratio. Calculate the χ^2 value by filling in the table below. [1]

Formula for χ^2 calculation

$$\chi^2 = \sum \frac{(O-E)^2}{E} \quad v = c - 1$$

where Σ = 'sum of...'
 v = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

Category	Observed number (O)	Expected ratio	Expected number (E)	$(O - E)^2$	$\frac{(O - E)^2}{E}$
finely striped leaves and high resistance					
finely striped leaves and low resistance					
entirely green leaves and high resistance					
entirely green leaves and low resistance					
TOTAL					

(ii)

Table 6

degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.31	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

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Examiner's
use

Using **Table 6**, explain the conclusion drawn from the Chi square test to determine if the observed numbers conformed to the expected. [3]

(iii) Explain the main reason behind the discrepancy between the actual and expected results. [1]

- (iv) Draw a genetic diagram to explain the difference between expected and actual results of the cross. [3]

Let **G** represent the allele for entirely green leaf.
Let **g** represent the allele for finely striped leaf.
Let **R** represent the allele for low resistance.
Let **r** represent the allele for high resistance;

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Total: [13]

Question 7

- (a) Myasthenia gravis is a disease which causes muscular weakness. It develops because of an attack by the body's own immune system on neuromuscular junctions. **Fig7.1** below shows a normal neuromuscular junction (nerve to muscle) and one affected by the disease (myasthenic).

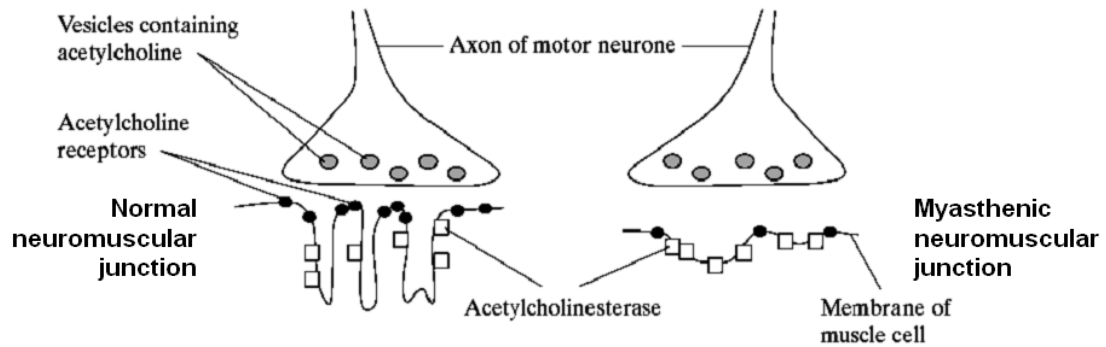


Fig7.1

Describe **two** ways in which a myasthenic neuromuscular junction differs from a normal one and explain how each difference would affect transmissions across the myasthenic neuromuscular junction. [4]

1.

2.

- (b) A myelinated axon uses less ATP to transmit a nerve impulse than an unmyelinated axon of the same diameter. Explain why. [1]

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Fig 7.2 shows a signaling pathway downstream of Ras. This pathway consists of a linear series of kinases that form a so-called MAPkinase (MAPK) cascade. **A**, **B**, **C** and **D** are the main processes in the signaling pathway.

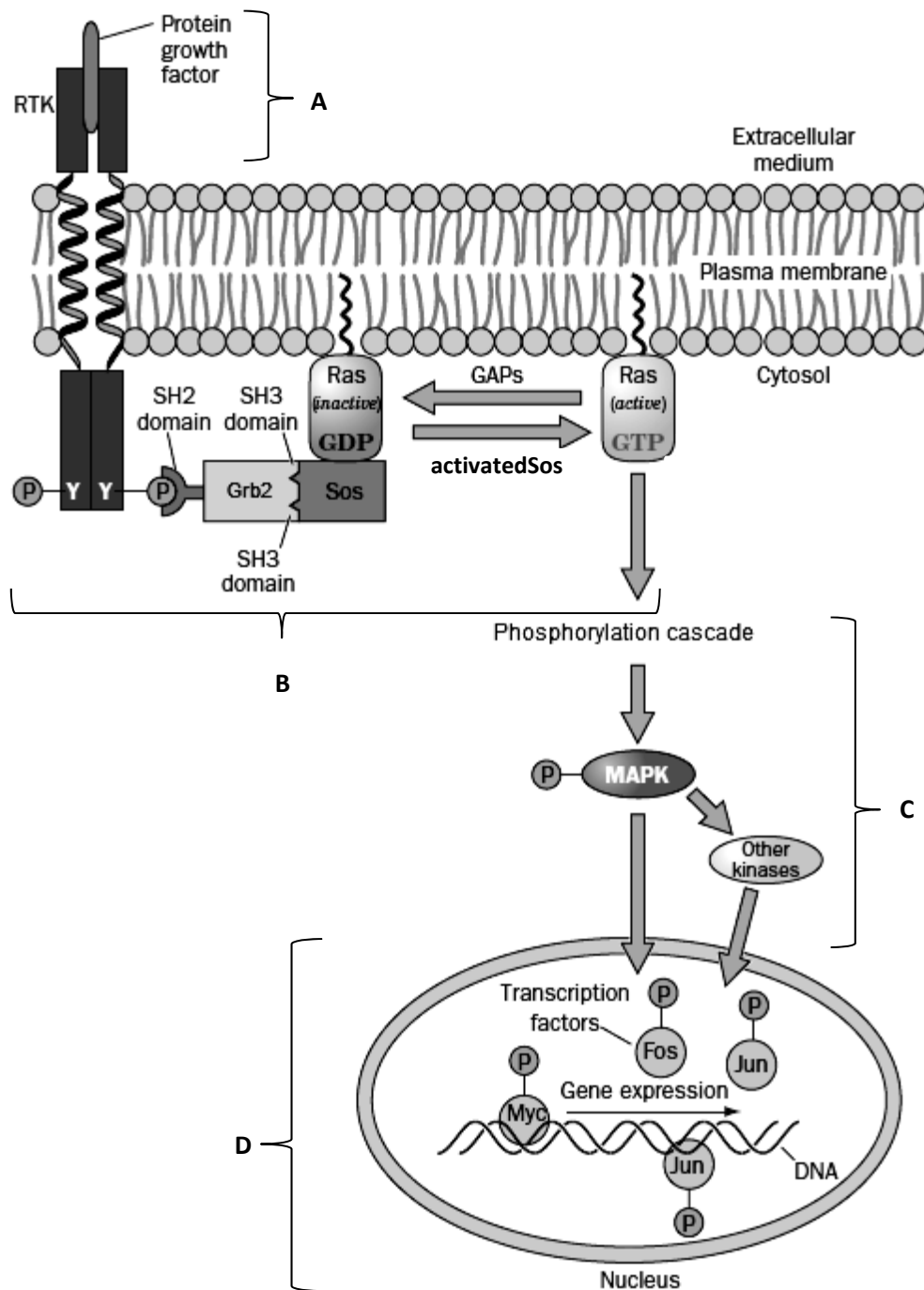


Fig 7.2

(c) Name the stages **A** and **D**. [1]

A: _____

D: _____

- (d) Based on your knowledge of tyrosine kinase receptor and **Figure 7.2**, describe processes **A to D**. [4]

*For
Examiner's
use*

Total: [10]

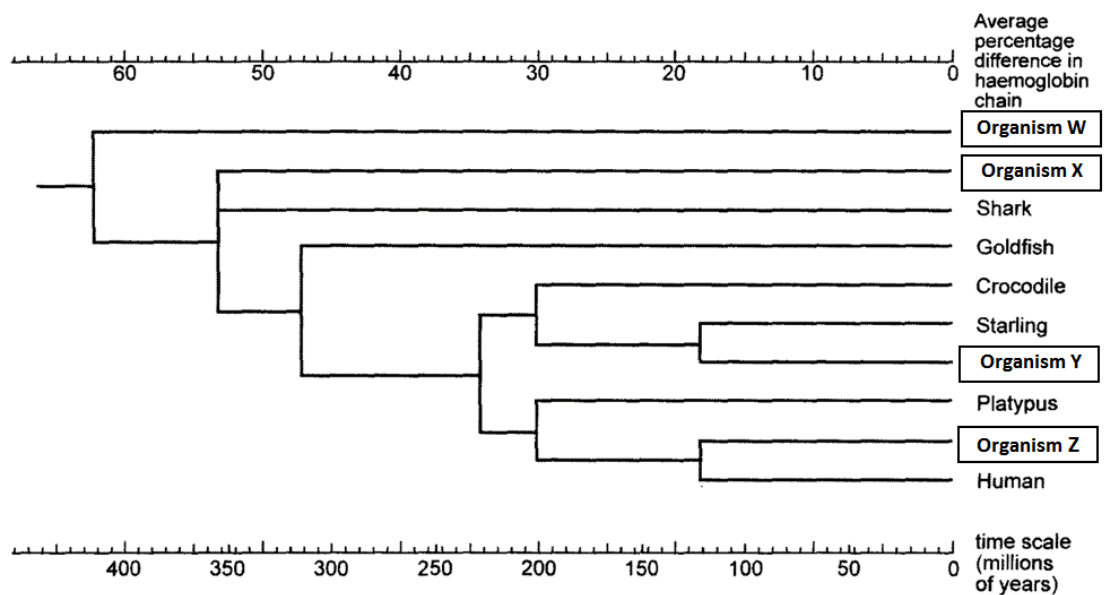
Question 8

- (a) **Table 8** shows the amino acid differences in the alpha haemoglobin chains between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Table 8

Fig 8 shows a family tree based on differences between the alpha haemoglobins in **Table 8**.

**Fig 8**

- (i) Using information from **Table 8**, identify organisms **W** to **Z**. [2]

W _____

X _____

Y _____

Z _____

- (ii) Explain how differences in amino acid sequences in the alpha haemoglobin chain allows the establishment of a phylogenetic tree. [3]

- (b) Explain the difference between classification and phylogeny. [2]

- (c) (i) State two advantages of using molecular evidence compared to homology in establishing phylogeny. [2]

- (ii) Suggest why homology still features prominently in evolutionary studies despite the advantages molecular evidence can confer. [1]

Total: [10]

Section C: Free-Response Question(20 marks)

Answer only **one** question.

Write your answers on the writing paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL RETURN** is required.

Question 1

- (a) Describe the structural features and negative regulation of the lac operon. [8]
- (b) Distinguish between repressible and inducible systems of gene regulation. [6]
- (c) Describe how horizontal gene transfer can occur in prokaryotes. [6]

OR

Question 2

- (a) Distinguish between the processes of Krebs Cycle and Calvin Cycle. [7]
- (b) Describe the role of oxidative phosphorylation. [7]
- (c) Discuss the effects of varying light intensity on photosynthesis. [6]

Total: [20]

END OF PAPER



**DUNMAN HIGH SCHOOL
PRELIMINARY EXAMINATION 2014
YEAR SIX
H2 BIOLOGY (9648)**

Suggested Answers:

1(a)

Protein C is a carrier protein / pump involved in active transport which transport substances against a concentration gradient

Binds & releases specific substances / desired molecules by changing the shape of the protein using ATP

(b)(i)

Phagocytosis

uptake of particulate material involving pseudopodia and engulfing of material

Formation of vesicles around the material involves the fusion of cell surface membrane as it pinched off;

(ii)

primary lysosomes containing hydrolytic enzymes (e.g. lipases, proteases, carbohydrases, nucleases)

fuse with vacuoles containing engulfed material / phagosome forming secondary lysosome; subsequent breakdown by hydrolysis of that material to soluble products

(c)

Quote: At 15 seconds, volume of oxygen produced for 25°C is 60cm³, twice that of 15°C, which is only 30cm³

As temperature increases from 15°C to 25°C, kinetic energy increases, frequency of effective collision increases, rate of enzyme-substrate complex formation increases

Rate of reaction increases as more products are formed per unit time

The rate of catalase activity follows Q₁₀, temperature coefficient is 2, where rate of reaction doubles for every increase in 10°C until optimal temperature

After 25 seconds, temperature is no longer has an effect on the activity of catalase / not the limiting factor, substrate concentration is limiting

(d)

A part of Atorvastatin is structurally similar / similar configuration to the substrate HMG-CoA

The drug is a competitive inhibitor which competes with HMG-CoA to bind the active site of the enzyme HMG-CoA reductase, reducing its activity

(e)

Hydrogen bonds are formed between C=O and N-H of the peptide bonds of the secondary structure α-helix and β-pleated sheets

Hydrogen, ionic, disulphide linkages formed between R-groups of the different polypeptide chains

(a)

B, A, C, D

(b)

During prophase, nuclear envelope starts to fragment and disintegrate

During telophase, the nuclear envelope re-forms around the chromosomes at each pole

(c)

Paclitaxel prevents the kinetochore microtubules from shortening during anaphase as a result the sister chromatids cannot be pulled to the opposite poles of the cell and mitosis cannot proceed

This prevents proliferation (rapid cell division) / multiplication of cells through mitosis of cancerous cells

(a)(i)

Nucleus

(ii)

TFIID/ TFIIA (general transcription factors) unable to recognise and bind promoter region; RNA polymerase unable to recognize and bind to promoter, transcription initiation complex cannot be formed

Transcription cannot occur and protein synthesis will be decreased

(b)(i)

5' GCUACGUCCUUGUACACAUU 3'

(ii)

RNA polymerase reads the DNA template strand in a 3' to 5' direction

Newly synthesised RNA is added based on complementary base pairing, A:U, T:A and C:G

(c)

	Transcription	Translation
Location	nucleus	cytoplasm
template	DNA template strand	mRNA
products	mRNA, rRNA, tRNA;	Polypeptide / protein;
bonds formed between	phosphodiester bond is formed between monomers;	peptide bond is formed between;
monomers	Free ribonucleotide triphosphate / ribonucleotides;	amino acids;
Enzyme	RNA polymerase;	Peptidyl transferase;

4(a)(i)

A: Hemagglutinin

B: Viral RNA / ssRNA

(ii)

Viral RNA contains RNA segment which acts as templates that is transcribed into viral proteins essential for the virus to bind to and infect cells

For example, glycoprotein hemagglutinin is involved in the specific attachment to the receptors found on epithelial cells

For example, enzyme neuraminidase is involved in the cleavage of sialic acid from the host cell receptors for the release of viral progeny

For example, RNA polymerase is involved in the synthesis of progeny viral RNA or viral proteins

(b)

They bind to different receptors, HIV glycoprotein 120 binds to CD4 receptors while influenza HA binds to receptors with sialic acid

For influenza virus, a low pH results in the fusion of the viral envelope and the endosome, releasing the viral RNA into the cytoplasm. For HIV virus, the viral envelope fuses with the cell surface membrane, releasing the viral RNA

For HIV virus, reverse transcription occurs, where the viral RNA is reverse transcribed to form cDNA. This does not occur in influenza virus, the viral RNA enters the nucleus after being released from the endosome

(c)

Influenza glycoproteins are first attached on the outside of host cell membrane

Near the membrane, there is assembly of capsomere proteins, viral genome and viral glycoproteins into virions

Host cell surface membrane then surrounds the viral RNA and protein and the completed new virion buds off from the host cell

5(a)

Category	Prokaryotic cell	Eukaryotic cell
Circular or linear	Each chromosome consists of circular DNA.	Each chromosome consists of long and linear DNA.
Presence of introns	DNA does not contain introns.	DNA contains introns.
Other regulatory sequences	There are less regulatory sequences, only the promoter, absence of introns or silencer.	There are more regulatory sequences such as introns, enhancer and silencer.
Origin of Replication	There is only one single origin of replication in prokaryotes.	There are multiple origins of replication in eukaryotes.
Centromere	No centromeres or telomeres are present in prokaryotes.	Centromeres and telomeres are present in eukaryotes.
Packing of DNA	There is no packing of DNA using histone proteins.	There are packing of DNA using histones proteins.
Size	Length of chromosome shorter compared to eukaryotes.	Length of the chromosome is longer compared to prokaryotes.
Location	Genetic material found in nucleoid region, not bound by nuclear envelope.	Genetic material found in nucleus, bound by nuclear envelope.
Similarity	Genetic material is double stranded DNA	

(b)(i)

5' cap of the mRNA serves as a binding site for eIF4F

increases affinity of binding of ribosome to the mRNA / promotes formation of translation initiation complex and results in translation of the mRNA

(ii)

PABP binds to polyA tail of mRNA which increases the stability of the mRNA

The half-life of the mRNA is increased and the mRNA remains in the cell longer for more translation into proteins

6(a)

Parental phenotype : White X White ;
 Parental genotype : AaBb X AaBb ;

Gametes formed : AB Ab aB ab X AB Ab aB ab

Punnett square:

	AB	Ab	aB	ab
AB	AABB	AABb	AaBB	AaBb
Ab	AABb	AAbb	AaBb	Aabb
aB	AaBB	AaBb	aaBB	aaBb
ab	AaBb	Aabb	aaBb	aabb

F₁ genotype: AABB, AaBB, AABb, Aabb, AAbb
 AaBb, aaBb, aaBB, aabb

F₁ phenotype: White Yellow ;

F₁ phenotypic ratio: 13 : 3 ;

(b)

Allele A codes for a functional enzyme that synthesizes the yellow pigment while allele a does not;

Allele B codes for a protein that inhibits the enzyme coded by A that synthesizes yellow pigment or it codes for a protein that /breaks down yellow pigment;

(c)(i)

Category	Observed number (O)	Expected ratio	Expected number (E)	$(O - E)^2$	$\frac{(O - E)^2}{E}$
finely striped leaves and high resistance	80	1	50	900	18
finely striped leaves and low resistance	20	1	50	900	18
entirely green leaves and high resistance	22	1	50	784	15.68
entirely green leaves and low resistance	78	1	50	784	15.68;
TOTAL	200				67.36 (2 d.p.);

(ii)

Since the calculated χ^2 value 67.36 > critical χ^2 value 7.82 at $p = 0.05$ at d.f.= 3, null hypothesis is rejected and alternative hypothesis is accepted.;

At 3 degree of freedom, the calculated value of χ^2 (148.32) lies above 16.27 in the chi square table, therefore the corresponding probability is $p < 0.001$.;

Conclusion:

The results of the χ^2 test suggest that there is a significant difference between the observed and the expected values at the 5% level. Any difference is not due to chance alone but other factors are at work. Thus we conclude that the data does not follow the ratio of 1:1:1:1.;

(iii)

The loci/genes for leaf colour/pattern and resistance are linked/on same chromosome (Accounts for larger number of parental phenotypes (78+92) finely striped & resistance + entirely green leaves & low resistance);

or

Crossing over between linked genes in meiosis 1 gives smaller number of recombinants phenotypes;

(Smaller number recombinants phenotypes are finely striped leaves and low resistance + entirely green leaves & resistant);

1 Max

(iv)

F₁ phenotype : entirely green leaves and low resistance x finely striped leaves and high resistance

F₁ genotype (test cross) : $\frac{G R}{g r}$ x $\frac{g r}{g r}$;

Gametes formed : $\frac{GR}{}$ $\frac{gr}{}$ $\frac{Gr}{}$ $\frac{gR}{}$ x $\frac{gr}{}$

Punnett square

	$\frac{GR}{}$	$\frac{gR}{}$	$\frac{Gr}{}$	$\frac{gr}{}$
$\frac{gr}{}$	$\frac{G R}{g r}$	$\frac{g R}{g r}$	$\frac{G r}{g r}$	$\frac{g r}{g r}$

 ;

F₂ Genotype : $\frac{G R}{g r}$ $\frac{g R}{g r}$ $\frac{G r}{g r}$ $\frac{g r}{g r}$

F₂ Phenotype : entirely green, low resistance striped leaves, low resistance entirely green, high resistance striped leaves, high resistance ;

7(a)

Difference 1: myasthenic has fewer folds/ less surface area / fewer receptors can be found;
Effect 1: due to fewer Na^+ ligand gated channels open thus less Na^+ influx, so take a longer time for depolarization to occur compared to normal

Difference 2:

myasthenic has acetylcholinesterase in shallower folds / more exposed;
or

myasthenic has lower ratio of receptors to acetylcholinesterase;

Effect 2: so acetylcholine are degraded / destroyed/hydrolysed before binding to acetylcholine receptors or faster degradation of transmitter

Difference 3: myasthenic has wider synaptic cleft;

Effect 3: so takes longer for neurotransmitters to diffuse across;

(b)

ATP required for active transport of Na^+ ions which are actively moved out only at nodes in myelinated / Na^+ (actively) moved out along whole length of axon in non-myelinated

(c)

A: ligand-receptor interaction;

D: cellular responses;

(d)

Protein growth factor binding to receptor of RTK induces the autophosphorylation / cross-phosphorylation of the tyrosine residues in the RTK

Grb2 binds to the resulting phosphorylated Tyrosine residues via its SH2 domain and simultaneously binds to Sos via its two SH3 domains

This activate Sos which subsequently binds to inactive Ras

As a result, Ras is activated when it exchanges its bound GDP for GTP

The phosphorylated MAPK then migrates to the nucleus, where it phosphorylates transcription factors such as Fos, Jun, and Myc, thereby modulating / regulate gene expression

8(a)(i)

W: lungfish

X: coelacanth;

Y: ostrich

Z: elephant;

(ii)

% of aa difference in Hb chain indicates relatedness / few difference indicates recent common ancestor / large difference indicates early divergence

based on constant rate of neutral mutations acting as molecular clock

provides quantitative data to construct phylogenetic tree

(b)

classification refers to grouping organisms based on similar characteristics

characteristics may be analogous and not homologous

phylogeny involves grouping organisms based on evolutionary relationship

similarity is due to inheritance from common ancestry

(c)(i)

Objective / quantifiable

Open to statistical analysis

Use of phenotypically non-visible characteristics / considers changes caused by silent mutation which is not shown on the phenotype

Based on the idea of molecular clock, that the rate of mutation is constant, species can be arranged in order of time of evolution

Morphological similarities may be due to convergence if two species are subjected to similar selective pressure but are from two different ancestral lineages

Compute small changes

(c)(ii)

easier to use as it often requires observation rather than machines

DNA / protein not always available e.g. in fossils

Essay Answers

1(a)

Marking Point	Answer
1	The lac operon has a promoter which is the RNA polymerase binding site;
2	Operator which is the binding site for repressor/inhibitor protein ;
3	three structural genes which are clustered together and controlled by one promoter;
4	Lac Z which codes for beta-galactosidase, Lac Y which codes for permease, Lac A which codes for transacetylase;
5	Lac I, coding for repressor/inhibitor protein;
6	cap (catabolic activator protein) binding site CAP site;

Marking Point	Answer
7	Negative regulation of lac operon involves the use of a repressor protein which inhibits transcription;
8	In absence of lactose, repressor protein blocks access of RNA polymerase to promoter;
9	RNA polymerase cannot transcribe the structural genes of lac operon;
10	Inducer allolactose binds to the repressor protein, causing it to change conformation and unable to bind to operator, inactivating it;
11	results in RNA polymerase able to bind and transcribe the lac operon, allowing the cell to breakdown lactose;

(b)

Marking Point	Category	Repressible	Inducible
12	Example	Trp operon	Basal transcription of Lac operon
13	Default state	ON, actively transcribed	OFF, not actively transcribed
14	Nature of effector molecule	Co-repressor trp	Inducer allolactose
15	Effect of effector	Decrease transcription	Increase transcription
16	Repressor coded by	Structural gene	Regulatory gene
17	Repressor synthesised	Inactive form	Active form
18	Ana / Catabolic	Anabolic, involved in synthesis of products	Catabolic, involved in breakdown of products
19	Active Repressor binds	Operator of operon	

(c)

Marking Point	Answer
20	Horizontal gene transfer involves exchange of genetic material between members of the same species or different species;
21	Transformation involves bacteria taking up of naked DNA from the environment and integrating it into its genome;
22	Cells which are competent are able to undergo transformation;
23	In conjugation, DNA is transferred between two prokaryotic cells that are temporarily joined. In <i>E.coli</i> , the F plasmid is involved;
24	The F plasmid codes for a sex pili to allow the F plasmid to be transferred;
25	The outcome is that a F ⁺ cell converts an F ⁻ cell to F ⁺ after transferring the F plasmid; 2 max for conjugation
26	Transduction involves bacteriophages carrying prokaryotic genes from one host cell to another;
27	transduction results from accidents that occur during the phage replicative cycle, resulting in a phage that carries host cell DNA;
28	Some of the host cell DNA is then incorporated into another recipient cell's chromosome by DNA recombination, a recombinant cell is formed; 2 max for transduction

2(a)

Marking Point		Krebs cycle	Calvin cycle
29	Location	Mitochondrial matrix	Chloroplast stroma
30	Substrate	Acetyl-CoA and oxaloacetate combines to form citrate	CO ₂ and Ribulose biphosphate (RuBP)
31	Products	Each glucose molecule gives rise to: 6 NADH 2 FADH ₂ 2 ATP 4 CO ₂	For every 3 molecules of CO ₂ that enter the cycle, one triose phosphate / G3P is made
32	Regenerated / Starting material	Oxaloacetate is the starting material that is eventually regenerated	Ribulose biphosphate (RuBP) is the starting material that is eventually regenerated
33	ATP	Produced via substrate level phosphorylation	Used in reduction of glycerate-3-phosphate where energy is required through hydrolysis of ATP
34	Electron carriers	Use NAD ⁺ and FAD for the	Uses NADPH / reduced

	/ donors	oxidation of the intermediates of the cycle by serving as electron acceptors	NADP ⁺ to reduce glycerate-3-phosphate to triose phosphate by serving as electron donors
35	Overall	Catabolic	Anabolic
36	Role of CO ₂	CO ₂ is released as a result of decarboxylation reactions	Required for carbon fixation. CO ₂ is used to convert Ribulose biphosphate (RuBP) to form an unstable 6C compound that breaks down to form glycerate-3-phosphate
37	Role of O ₂	Occurs only when O ₂ is present	Does not require O ₂

(b)

Marking Point	Answer
38	Oxidative phosphorylation is the synthesis of ATP in the presence of oxygen and takes place within the inner mitochondrial membrane;
39	Reduced NAD and reduced FAD provide high energy electrons for the synthesis of ATP;
40	Electrons are passed along the Electron Transport Chain (ETC) from one electron carrier to the next, each with an energy level lower than the one preceding it;
41	As the electrons are passed from one electron carrier to the next, some of their energy is used to transport protons from matrix of the mitochondrion into the intermembrane space;
42	This produces a high concentration of H ⁺ in the intermembrane space, setting up an electrochemical proton gradient / proton motive force;
43	Stalk particles containing ATP synthases are embedded on inner mitochondria membrane;
44	H ⁺ diffuse through them, down their electrochemical gradients, back into the matrix;
45	This provides enough energy to synthesise ATP by the phosphorylation of ADP with inorganic phosphate;
46	Oxygen functions as the final proton and electron acceptor to form water, catalyzed by cytochrome oxidase;
47	NAD ⁺ and FAD are regenerated at the end of oxidative phosphorylation;

(c)

Marking	Answer
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Point	
48	Draw curve / graph of effects of varying light intensity on rate of photosynthesis + label light saturation point)
49	Light intensity directly affects light-dependent stage of photosynthesis. When light energy is absorbed by chlorophyll, electrons are excited to a higher energy level to be passed down the electron transport chain.
50	The process of photophosphorylation is responsible for both the photolysis of water to produce O_2 and the synthesis of ATP and NADPH to be used for Calvin Cycle where CO_2 is taken up.
50a	The rate of photosynthesis is often measured by the amount of CO_2 absorbed or O_2 evolved by the plant.
51	At lower light intensities, light intensity is the limiting factor where rate of photosynthesis increases proportionally with increasing light intensity
52	With increasing light intensity, compensation point is reached where the amount of carbon dioxide produced in respiration exactly balances that being used in photosynthesis
53	Further increase in light intensity results in the light saturation point being reached where light intensity will have no effect on the rate of photosynthesis
54	Light intensity is no longer a limiting factor, other environmental factors like CO_2 conc or temperature may be limiting now.
55	At very high light intensity, chlorophyll may be bleached (a condition called chlorosis), thus retarding photosynthesis.